9.4 Polarization of Wave :

Polarization of a wave refers to the time varying behavior of the electric field strength vector at some fixed point in space.

Let us consider the wave travelling in +z direction and hence it has E_x and E_y components as :

$$E_x = E_1 \cos(\omega t - \beta z)$$
$$E_y = E_2 \cos(\omega t - \beta z + \delta)$$

where, E_1 = amplitude of wave in x direction,
E_2 = amplitude of wave in y direction,
δ = Phase angle between E_x and E_y

These field components are shown in Fig. 9.4.1.

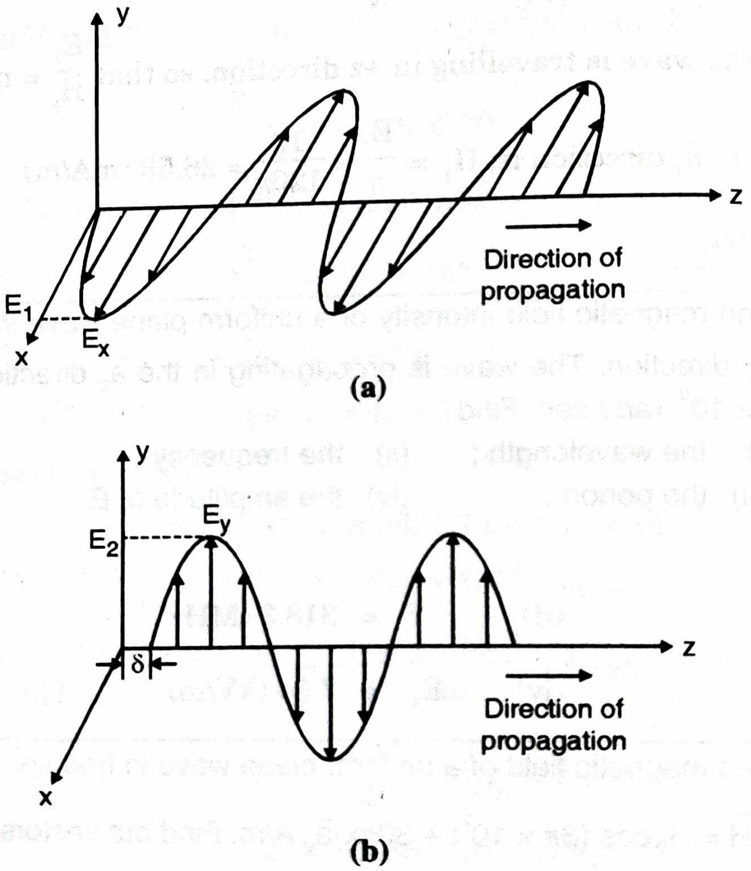

Fig 9.4.1 : Showing in E_x and E_y components for the wave in z-direction

Now we can explain polarization. There are three types of polarization.

- Linear polarization
- Elliptical polarization and
- Circular polarization

9.4.1 Linear Polarization :

A time-harmonic field is linearly polarized at a given point in space if the electric field vector at that point is always oriented along the same straight line at every instant of time.

The linear polarization is possible if the electric field vector possess,

- only one field component or
- two orthogonal linearly polarized components that are in time phase or 180° out of phase.

(a) Consider for a wave only E_x component is present and $E_y = 0$, Fig. 9.4.1(a).

i.e.
$$\bar{E} = E_1 \cos(\omega t - \beta z)\, \bar{a}_x$$

The variation of wave is shown in Fig. 9.4.2(a), which is obtained by observing the Fig. 9.4.1(a) from right. We see a point moving horizontally. This variation clearly shows that E is linear.

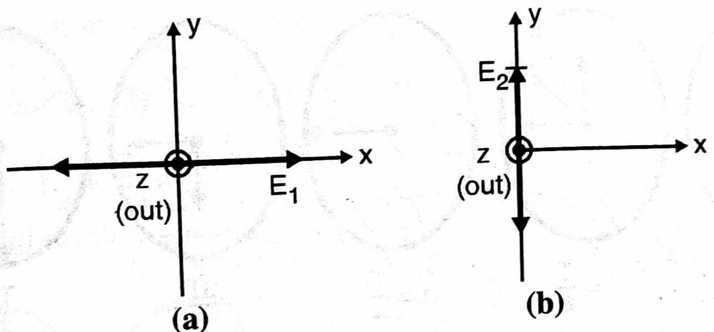

Fig. 9.4.2 : Showing linear polarization

In this case the wave is called as linearly polarized in x direction.

(b) If E has only E_y component and $E_x = 0$. (Fig. 9.4.1(b))

i.e.
$$\bar{E} = E_2 \cos(\omega t - \beta z + \delta)\, \bar{a}_y$$

The variation is shown in Fig. 9.4.2(b) obtained by observing Fig. 9.4.1(b), from right. The wave is called as linearly polarized in y direction.

(c) If both E_x and E_y are present and are in phase, the resultant electric field has a direction dependent on the relative magnitude of E_x and E_y. The angle this direction made by with the x axis is $\tan^{-1}(E_y / E_x)$ and this angle is always constant with time. This is shown in Fig. 9.4.2(c).

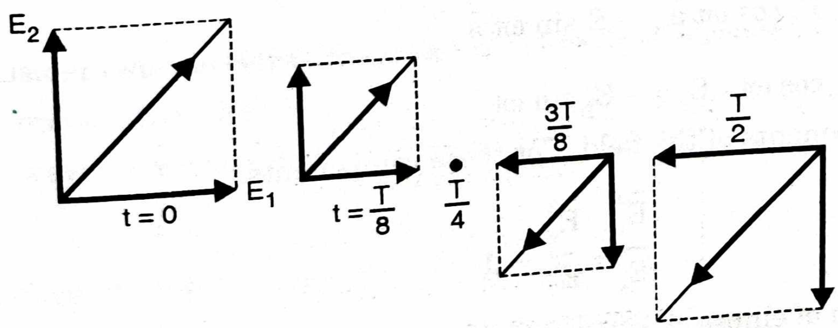

Fig. 9.4.2(c) : Showing linear polarization

In all the above cases i.e. (a), (b) and (c) the direction of the resultant vector is constant with time and the wave is said to be linearly polarized.

9.4.2 Elliptical Polarization :

A wave is said to be elliptically polarized if the tip of the electric field vector traces an elliptical locus in space. At various instants of time the electric field vector changes continuously with time in such a manner as to describe an elliptical locus.

It is **right-hand** elliptically polarized if the electric field vector of the ellipse rotates **clockwise** when viewed along the axis of propagation.

It is **left-hand** elliptically polarized if the electric field vector of the ellipse rotates **counterclockwise.**

Consider a case with $E_1 \neq E_2$ and $\delta = 90°$

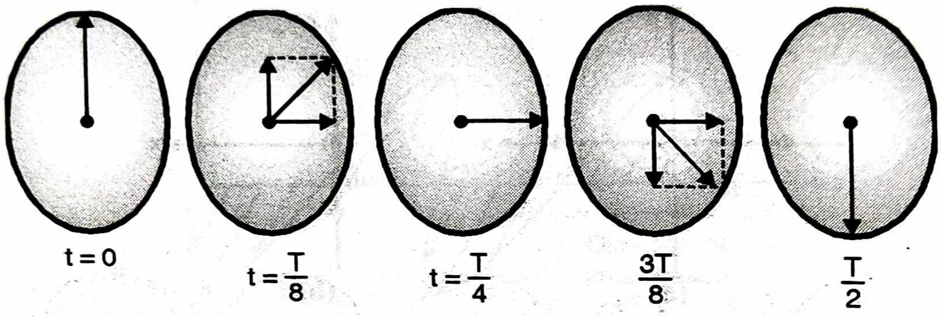

Fig. 9.4.3 : Showing elliptical polarization

The total electric field is a complex vector and is given by

$$E_0 = E_x \bar{a}_x + jE_y \bar{a}_y \qquad \text{...(at } z = 0)$$

The j term indicates that y component leads the x component by 90° and E_x and E_y are positive real constants.

Now the time varying field is

$$\tilde{E}(0,t) = \mathrm{Re}(E_x \bar{a}_x + jE_y \bar{a}_y)e^{j\omega t}$$

$$= \mathrm{Re}\{(E_x \bar{a}_x + jE_y \bar{a}_y)(\cos \omega t + j\sin \omega t)\}$$

$$= E_x \cos \omega t\, \bar{a}_x - E_y \sin \omega t\, \bar{a}_y$$

Let $\tilde{E}_x = E_x \cos \omega t \,;\ \tilde{E}_y = -E_y \sin \omega t$

These are the components of the field. For these components.

$$\frac{\tilde{E}_x^2}{E_x^2} + \frac{\tilde{E}_y^2}{E_y^2} = 1$$

This is the equation of ellipse.

9.4.3 Circular Polarization :

A wave is said to be circularly polarized if the tip of the electric field vector traces out a circular locus in space.

At various instants of time, the electric field intensity of such a wave always has the same amplitude and the orientation in space of the electric field vector changes continuously with time in such a manner as to describe a circular locus.

Consider a case with $E_1 = E_2 = A$ and $\delta = 90°$

$$E_0 = (\bar{a}_x + j\bar{a}_y)A \quad \text{...(at } z = 0)$$

Now the time varying field is

$$\tilde{E}(0,t) = \text{Re}\{(A\bar{a}_x + jA\bar{a}_y)e^{j\omega t}\} = (\bar{a}_x \cos\omega t - \bar{a}_y \sin\omega t)A$$

Thus, the x and y components of the field are

$$\tilde{E}_x = A\cos\omega t \; ; \; \tilde{E}_y = A\sin\omega t$$

For these components $\tilde{E}_x^2 + \tilde{E}_y^2 = A^2$

This is the equation of circle with radius A.

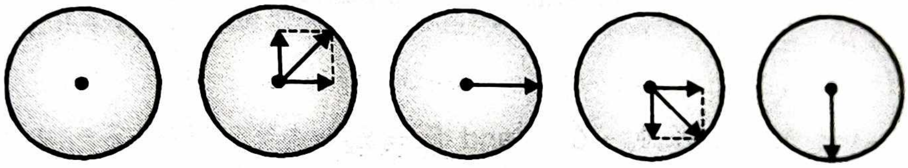

Fig. 9.4.4 : Showing circular polarization

When the tip of the field vector rotates with time in the clockwise sense as seen looking along the direction of propagation of the wave, the wave is said to be right polarized. When the electric vector has a counter clockwise sense of rotation when it is viewed along the axis of propagation, then the wave is said to be left polarized.

9.4.4 Relation between Linear and Circular Polarization :

Let us consider two waves left and right circulatory polarized of the same amplitude, frequency and propagation direction, and having some phase shift δ radians between them. Say, the waves are travelling in + z direction. The field expression for these two waves are,

For left circulatory polarized → $E_L = E_0(\bar{a}_x + j\,\bar{a}_y)\,e^{-j\beta z}\,e^{j\delta}$

For right circulatory polarized → $E_R = E_0(\bar{a}_x - j\,\bar{a}_y)\,e^{-j\beta z}$

The addition of these two waves

$$E_T = E_0 (\bar{a}_x - j\,\bar{a}_y)\, e^{-j\beta z} + E_0 (\bar{a}_x + j\,\bar{a}_y)\, e^{-j\beta z}\, e^{j\delta}$$

$$= E_0 \left[(1 + e^{j\delta})\, \bar{a}_x - j\,(1 - e^{j\delta})\, \bar{a}_y \right] e^{-j\beta z}$$

$$= E_0\, e^{j\delta/2} \left[(e^{-j\delta/2} + e^{j\delta/2})\, \bar{a}_x\, j\, (e^{-j\delta/2} - e^{j\delta/2})\, \bar{a}_y \right] e^{-j\beta z}$$

Using, $\quad e^{j\delta/2} + e^{-j\delta/2} = 2\cos(\delta/2)$

and $\quad e^{j\delta/2} - e^{-j\delta/2} = 2j\sin(\delta/2)$

$$E_T = E_0 \left[2\cos(\delta/2)\, \bar{a}_x - j\,(2j\sin\delta/2)\, \bar{a}_y \right] e^{-j\beta z}\, e^{j\delta/2}$$

$$= 2E_0 \left[\cos(\delta/2)\, \bar{a}_x + \sin(\delta/2)\, \bar{a}_y \right] e^{-j(\beta z - \delta/2)}$$

This represents the electric field of a linearly polarized wave.

Conclusion : Two circularly polarized waves addition gives a linearly polarized wave.

Example 9.4.1 : Consider the instantaneous electric field is given by,

$$\bar{E} = E_1 \cos(\omega t - \beta z + \phi_x)\, \bar{a}_x + E_2 \cos(\omega t - \beta z + \phi_y)\, \bar{a}_y$$

Determine the polarization in z = 0 plane if

(i) $E_1 = 0$

(ii) $\phi_x = \phi_y = \phi$

(iii) $\phi_x = +\pi/2, \phi_y = 0$ and $E_1 = E_2 = A$

(iv) $\phi_x = 0, \quad \phi_y = \frac{\pi}{2}$ and $E_1 = E_2 = B$

(v) $\phi_x = +\pi/2, \phi_y = 0$ and $E_1 = A$; $E_2 = B$

Solution : We observe these fields at some z = constant plane, e.g. z = 0 plane.

(i) $E_1 = 0$

The electric field for this case is

$$\bar{E} = E_2 \cos(\omega t + \phi_y)\, \bar{a}_y$$

The locus of the instantaneous electric field vector will be a straight line, and will be always directed along the y-axis at all times as shown in Fig. Ex. 9.4.1(a).

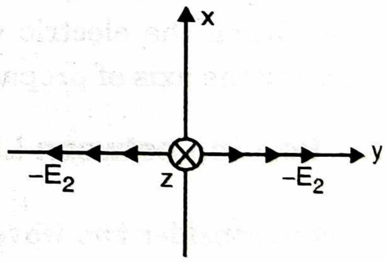

Fig. (a) : Illustrating Ex. 9.4.1(a)

(ii) $\phi_x = \phi_y = \phi$

In this case the electric field is given by,

$$\bar{E} = E_1 \cos(\omega t + \phi)\, \bar{a}_x + E_2 \cos(\omega t + \phi)\, \bar{a}_y$$

The amplitude of the electric field is

$$E = \sqrt{E_1^2 \cos^2(\omega t + \phi) + E_2^2 \cos^2(\omega t + \phi)} = \sqrt{E_1^2 + E_2^2}\, \cos(\omega t + \phi)$$

This is a straight line.

The direction of electric field is along the line that makes an angle θ with x-axis given by,

$$\theta = \tan^{-1}\left(\frac{E_2 \cos(\omega t + \phi)}{E_1 \cos(\omega t + \phi)}\right) = \tan^{-1}\left(\frac{E_2}{E_1}\right)$$

The electric field is thus linearly polarized along a line making angle θ with x-axis as shown in Fig. Ex. 9.4.1(b).

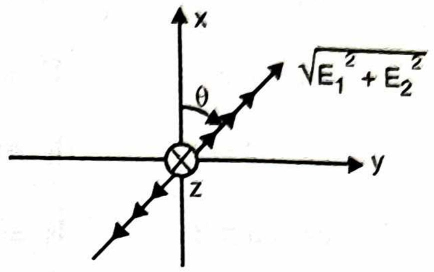

Fig. (b) : Illustrating Ex. 9.4.1(b)

(iii) $\phi_x = +\pi/2$, $\phi_y = 0$ and $E_1 = E_2 = A$

The instantaneous electric field is

$$\bar{E} = A\cos\left(\omega t + \frac{\pi}{2}\right)\bar{a}_x + A\cos(\omega t)\,\bar{a}_y$$

$$= -A\sin(\omega t)\,\bar{a}_x + A\cos(\omega t)\,\bar{a}_y$$

The amplitude of it is

$$E = \sqrt{(-A\sin\omega t)^2 + (A\cos\omega t)^2} = A$$

Which is constant irrespective of time. The angle θ w.r.t x axis is

$$\theta = \tan^{-1}\left(\frac{-A\cos\omega t}{A\sin\omega t}\right) = \tan^{-1}[-\cot\omega t] = \omega t + \frac{\pi}{2}$$

The locus of the field vector is a circle of radius A and it rotates clockwise with an angular frequency ω as shown in Fig. Ex. 9.4.1(c). Thus this is a left hand circular polarization.

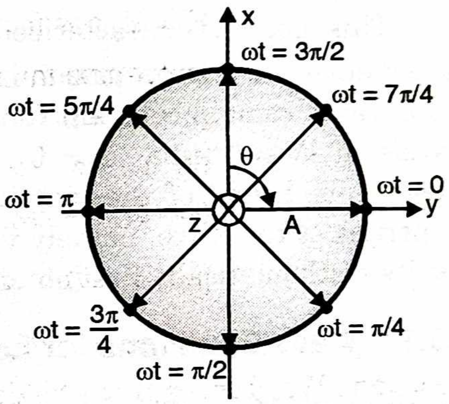

Fig. (c) : Illustrating Ex. 9.4.1(c)

(iv) $\phi_x = 0$, $\phi_y = \frac{\pi}{2}$ and $E_1 = E_2 = B$

The instantaneous electric field is given by

$$\bar{E} = B\cos(\omega t)\,\bar{a}_x + B\cos\left(\omega t + \frac{\pi}{2}\right)\bar{a}_y$$

$$= B\cos(\omega t)\,\bar{a}_x - B\sin(\omega t)\,\bar{a}_y$$

The amplitude is $E = \sqrt{(B\cos\omega t)^2 + (-B\sin\omega t)^2} = B$

Which is constant irrespective of time.

The angle θ ω.r.t. x axis is

$$\theta = \tan^{-1}\left(\frac{-B\sin(\omega t)}{B\cos(\omega t)}\right)$$

$$= \tan^{-1}[-\tan(\omega t)] = -\omega t$$

The locus of field vector is circle of radius B and it rotates anticlockwise with an angular frequency ω as shown in Fig. Ex. 9.4.1(d).

Thus this is a left hand circular polarization.

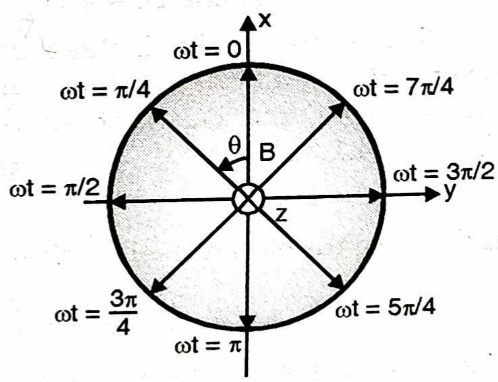

Fig. (d) : Illustrating Ex. 9.4.1(d)

(v) $\phi_x = +\pi/2, \phi_y = 0$ and $E_1 = A ; E_2 = B$

The instantaneous electric field is given by

$$\bar{E} = A\cos\left((\omega t) + \frac{\pi}{2}\right)\bar{a}_x + B\cos(\omega t)\,\bar{a}_y = -A\sin(\omega t)\,\bar{a}_x + B\cos(\omega t)\,\bar{a}_y$$

At different values of ωt we can plot $\bar{E}$ as shown below, Fig. Ex. 9.4.1(e)

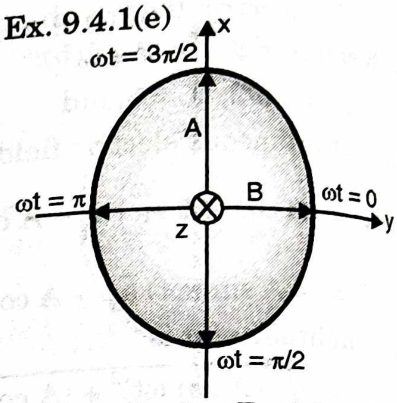

Fig. (e) : Illustrating Ex. 9.4.1(e)

at $\omega t = 0$ $\quad \bar{E} = B\,\bar{a}_y$

$\omega t = \pi/2$ $\quad \bar{E} = -A\,\bar{a}_x$

$\omega t = 2\pi/2$ $\quad \bar{E} = -B\,\bar{a}_y$

$\omega t = 3\pi/2$ $\quad \bar{E} = A\,\bar{a}_x$

and so on. As time progresses E rotates clockwise.

Thus it is a elliptical polarization (right hand).

9.5 Wave Propagation in Lossy Media :

This media is characterized by ($\sigma \neq 0, \epsilon = \epsilon_0 \epsilon_r, \mu = \mu_0 \mu_r$). In sections 9.2 and 9.3 we have studied wave propagation in lossless media. Due to zero loss ($\sigma = 0$), the amplitude of these waves remain constant (does not decrease). Then it is very obvious that when wave travels in lossy media ($\sigma \neq 0$), the energy of the wave should continuously decrease, causing amplitude of the wave decrease continuously. Except the amplitude, all other properties of the wave, namely v, λ . β are same as that for lossless media. To obtain the results mathematically first obtain wave equations for this media.

9.5.1 Wave Equations for Lossy Media :

Starting with Maxwell's equations,

$$\nabla \times \bar{H} = \bar{J} + \dot{\bar{D}} \quad \text{... (i)}$$

$$\nabla \times \bar{E} = -\dot{\bar{B}} \quad \text{... (ii)}$$

$$\nabla \cdot \bar{D} = 0 \quad \text{i.e } \nabla \cdot \bar{E} = 0 \quad \text{... (iii)}$$

$$\nabla \cdot \bar{B} = 0 \quad \text{i.e } \nabla \cdot \bar{H} = 0 \quad \text{... (iv)}$$

also

$$\bar{D} = \epsilon \bar{E} \quad \therefore \dot{\bar{D}} = \epsilon \dot{\bar{E}} \quad \text{... (v)}$$

$$\bar{B} = \mu \bar{H} \quad \therefore \dot{\bar{B}} = \mu \dot{\bar{H}} \quad \text{... (vi)}$$

$$\bar{J} = \sigma \bar{E} \quad \therefore \dot{\bar{J}} = \sigma \dot{\bar{E}} \quad \text{... (vii)}$$